

If no valid point x_i labeled 0 that satisfies the condition of the boundary is found, then we simply set $a_S = a_m$. The same thing goes for the rest of the points, resulting in a new indicator function $h'_S = h_{(a'_S, b'_S, c'_S, d'_S)} = \mathbb{1}_{[a'_S, b'_S] \times [c'_S, d'_S]}$

4.2 b) find the sample complexity $m_{\mathcal{H}_\alpha}(\epsilon, \delta)$ in order to show that $\mathcal{H}_\alpha$ is PAC - learnable.

5 Exercise 5.

Compute $\text{VCdim}(\mathcal{H})$ where $\mathcal{H}$ is the following hypothesis class:

$$\mathcal{H} = \{h_\theta : \mathbb{R} \rightarrow \{0, 1\} \mid h_\theta(x) = \mathbf{1}_{[\theta, \theta+1] \cup [\theta+2, \theta+4] \cup [\theta+6, \theta+9]}(x), \theta \in \mathbb{R}\}$$

From the beginning, we can conclude that the maximum dimension that $\mathcal{H}$ can have is 6. Taking a set of 7 elements would make it impossible for the label (1,0,1,0,1,0,1) to appear. Now we can look at the smaller dimensions. Also, considering that we have 3 intervals, it's safe to assume that the VC dimension of $\mathcal{H}$ is greater or equal to 2.

For the dimension of 3, we can continue with another simple set $\mathcal{C} = \{1, 2, 3\}$, alongside the theta values of:

- $\theta = 1000$ to achieve (0,0,0)
- $\theta = 3$ to achieve (0,0,1)
- $\theta = 1.5$ to achieve (0,1,0)
- $\theta = -8$ to achieve (1,0,0)
- $\theta = -7$ to achieve (1,1,0)
- $\theta = 2$ to achieve (0,1,1)
- $\theta = -3$ to achieve (1,0,1)
- $\theta = -5$ to achieve (1,1,1)

Thus, $\mathcal{C}$ can be shattered, resulting in $\text{VCdim}(\mathcal{H}) \geq 3$. We can also find a set of 4 points that can be shattered by h_θ . Consider $\mathcal{C} = \{1, 2, 6.5, 8.5\}$.

1.
 - If $\theta = -5.0 \rightarrow (1, 1, 0, 0)$
 - If $\theta = -4.9 \rightarrow (0, 1, 0, 0)$
 - If $\theta = -3.9 \rightarrow (0, 0, 0, 0)$
 - If $\theta = -3.0 \rightarrow (1, 0, 0, 0)$
 - If $\theta = -2.5 \rightarrow (1, 0, 1, 0)$
 - If $\theta = -2.0 \rightarrow (1, 1, 1, 0)$
 - If $\theta = -0.9 \rightarrow (0, 1, 1, 0)$
 - If $\theta = -0.5 \rightarrow (0, 1, 1, 1)$
 - If $\theta = -0.0 \rightarrow (1, 1, 1, 1)$
 - If $\theta = 0.1 \rightarrow (1, 0, 1, 1)$
 - If $\theta = 0.6 \rightarrow (1, 0, 0, 1)$
 - If $\theta = 1.0 \rightarrow (1, 1, 0, 1)$
 - If $\theta = 1.1 \rightarrow (0, 1, 0, 1)$
 - If $\theta = 2.1 \rightarrow (0, 0, 0, 1)$
 - If $\theta = 2.5 \rightarrow (0, 0, 1, 1)$
 - If $\theta = 2.6 \rightarrow (0, 0, 1, 0)$

Thus, we can consider that $\mathcal{C}$ is shatterable, resulting in $\text{VCdim}(\mathcal{H}) \geq 4$.

Now, finding 32 values for θ and 5 different values for x that would fit this general case, seems a little bit impossible. Let's see if we can prove that the VCdim of $\mathcal{H}$ is lower than 5.

Consider the general case $a < b < c < d < e$, and the subset 1 0 1 0 1. We want to place a in the first interval, b in none, c in the second interval, d in none, and e in the third interval. This ordering is based on the order of real numbers.

If we take a to be in $[\theta, \theta + 1]$, b not in this interval, and c in $[\theta + 2, \theta + 4]$, then the distance (a, b) must be strictly greater than 1. This is because the distance of 1 is between $\theta + 1$ and $\theta + 2$. The same argument applies to the intervals $[\theta + 4, \theta + 6]$ for c and e. Therefore, the distance from c to e is greater than 2, which is the minimum distance needed to have a gap between $\theta + 4$ and $\theta + 6$. Thus, the distance from a to e is strictly greater than 3.

We also note that the intervals have lengths 1, 2, and 3. Therefore, it is not possible to obtain all numbers in the same interval. This leads to a contradiction, which implies that the subset 1 1 1 1 1 cannot be obtained.

For the subset 1 0 1 0 0, we should have:

- a in $[\theta, \theta + 1]$, b in none, c in $[\theta + 2, \theta + 4]$, d and e in none
- a in $[\theta + 2, \theta + 4]$, b in none, c in $[\theta + 6, \theta + 9]$

To achieve the first situation, the distance from a to c must be at most 4, which is the extreme case where a is the left endpoint of the first interval and c is the

right endpoint of the second interval. To achieve the second situation, the distance from a to c must be at most $7 ((\theta + 9) - (\theta + 2))$. Thus, taking the less restrictive situation, we obtain that $\text{dist}(a, c) \leq 7$.

The idea is that at this rate, we should either arrive at a contradiction or a "recipe" that tells us exactly how to construct the set that is a good example.

We can also consider the symmetric case $0\ 0\ 1\ 0\ 1$, which gives:

- c in $[\theta, \theta + 1]$, d in none, e in $[\theta + 2, \theta + 4]$, a and b in none
- c in $[\theta + 2, \theta + 4]$, d in none, e in $[\theta + 6, \theta + 9]$, a and b in none

From here, we can obtain $\text{dist}(c, e)$ is at most 4 and at most 7, respectively, which implies that the total distance is at most 7.